Practical 3.1 & 3.2: KoboToolbox Installation and Form Design

Self-hosted installation with kobo-install and Tanga flood survey form

Day 3 · 3 hours · Resilience Academy SDI Training

Reference: `https://github.com/kobotoolbox/kobo-install` | Data: Tanga & Mwanza, Tanzania

Learning Objectives:

- Install a self-hosted KoboToolbox server using kobo-install
- Configure the server for local development (Workstation mode)
- Verify the installation and create an admin account
- Design a Tanga flood survey form with GPS coordinates

Prerequisites

- **Python 3.10+** installed
- **Git** installed and accessible from the terminal
- **Docker** with Compose V2 (V1 is NOT supported)
- At least 4GB RAM available
- Port 80 free (stop CRD if running: `cd /my_crd && docker compose down`)

Critical: The kobo-install script calls `git` internally to clone the `kobo-docker` repository. If git is not installed or not in PATH, you will get:
`FileNotFoundError: [Errno 2] No such file or directory: ''`
Fix: `sudo apt install -y git`

Step 1: Verify Prerequisites

```
# Verify all tools are available
git --version # must show git 2.x+
python3 --version # must show 3.10+
docker --version # must show Docker 24.x+
docker compose version # must show V2 (NOT V1)

# Free port 80 if CRD is running
cd ~/my_crd && docker compose down 2>/dev/null
cd ~
```

Step 2: Clone kobo-install

```
cd ~
git clone https://github.com/kobotoolbox/kobo-install.git
cd kobo-install
```

Step 3: Run the Setup Wizard

```
python3 run.py
```

The first time you run this, the interactive setup wizard launches. Answer the prompts:

```
What kind of installation? --> 1 (Workstation)
Installation directory? --> Accept default (../kobo-docker)
Use HTTPS? --> 2 (No)
Super user's username? --> admin
Super user's password? --> Admin123
Activate backups? --> 2 (No)
```

The wizard then:

1. Clones `kobo-docker` into `../kobo-docker`
2. Generates environment files in `.kobo/envfiles/`
3. Pulls Docker images from Docker Hub
4. Starts all containers

If you see `FileNotFoundError`: Git is not installed or the setup config is corrupted. Fix:

```
sudo apt install -y git
rm -f ~/kobo-install/.run.conf
cd ~/kobo-install && python3 run.py
```

Step 4: Verify and Monitor

```
# Check installation info
python3 run.py --info

# View live logs
python3 run.py --logs

# Access KoboToolbox:
# Main interface: http://localhost
# Django admin: http://localhost/admin
```

Step 5: Useful kobo-install Commands

```
python3 run.py # Start KoboToolbox
python3 run.py --stop # Stop all containers
python3 run.py --logs # View container logs
python3 run.py --info # Show installation info
python3 run.py --setup # Re-run setup wizard
python3 run.py --update # Update to latest version
python3 run.py --help # Show all commands
```

Step 6: Design the Tanga Flood Survey Form

In KoboToolbox web interface (`http://localhost`):

1. Click **New** → **Build from scratch**
2. Name: `Tanga Flood Survey 2026`
3. Add these questions:

```
Question: GPS Location | Type: Point (geopoint)
Question: Flood depth (cm) | Type: Integer
Question: Date observed | Type: Date
Question: Ward name | Type: Select one
  Options: Chumbageni, Makorora, Ngamiani,
          Mzingani, Tanganyika, Other
Question: Photo of site | Type: Photo
Question: Affected households | Type: Integer
Question: Damage type | Type: Select multiple
  Options: Road, Bridge, Building,
          Electricity, Water supply, None
Question: Observer notes | Type: Text area
```

Deploy the form:

1. Click **Save**
2. Click **Deploy**
3. Open Enketo web form (Preview)
4. Submit 3 test records with Tanga GPS coordinates (lat ≈ -5.07, lon ≈ 39.10)

Troubleshooting

```
# Port 80 already in use?
sudo lsof -i :80
# Stop whatever is using it, then retry

# Docker Compose V1 not supported?
docker compose version # must be V2
sudo apt remove docker-compose
sudo apt install docker-compose-plugin

# Need to restart from scratch?
python3 run.py --setup

# Containers crashing? Check memory:
docker stats
# Increase Docker memory to 4GB+ minimum
```

Checkpoint:

✓ `python3 run.py -info` shows all services running
✓ `http://localhost` loads the KoboToolbox login page
✓ Admin account login works
✓ Tanga flood survey form deployed successfully
✓ 3 test submissions visible in Data → Table view
✓ GPS coordinates visible in Data → Map view